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k-Nearest Neighbors

Distance metrics for classification

1. Euclidean distance (L2 norm)

$\vec{x_1}: (x_{11}, x_{21}, \ldots, x_{p1})$
$\vec{x_2}: (x_{12}, x_{22}, \ldots, x_{p2})$

$$\text{Euclidean distance} = \sqrt{\sum_{i=1}^{p} (x_{i1} - x_{i2})^2}$$

e.g. $x_1 = (1,2)$, $x_2 = (4,6)$
$d_{Euclidean} = \sqrt{3^2 + 4^2} = 5$

e.g. $x_1 = (1,1,2,3)$ $x_2 = (2,4,1,5)$
$d_{Euc} = \sqrt{1^2 + 2^2 + 1^2 + 2^2} = \sqrt{15}$

2. Manhattan distance (taxicab or L1 norm)

$$\text{Manhattan Distance} = \sum_{i=1}^{p} |x_{i1} - x_{i2}|$$

e.g. (1,2), (4,6)
$d = 3 + 4 = 7$

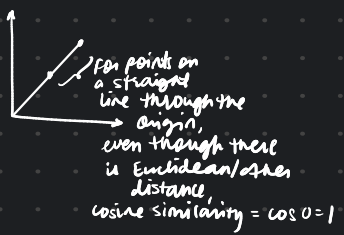
(1,1,2,3), (2,4,1,5)
$d = 1 + 3 + 1 + 2 = 7$

3. Minkowski distance (norm)

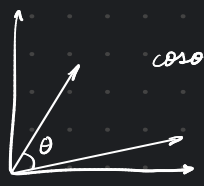
$$\text{Minkowski distance} = \left(\sum_{i=1}^{p} |x_{i1} - x_{i2}|^t \right)^{1/t}$$

$t = 1 \rightarrow$ Manhattan distance
$t = 2 \rightarrow$ Euclidean distance

e.g. (1,2), (4,6)
$d_{\text{minkc}(t=3)}(x_1, x_2) = (3^3 + 4^3)^{1/3} = 4.498$



4. Cosine similarity
↳ similarity is like the inverse of distance (more distance, less similar).



- used when the magnitude of the vectors doesn't matter (input features are normalized to the same scale, say 0 to 1)

$$\text{cosine similarity} = \frac{\sum_{i=1}^{p} x_{i1} \cdot x_{i2}}{\sqrt{\sum_{i=1}^{p} x_{i1}^2} \sqrt{\sum_{i=1}^{p} x_{i2}^2}}$$

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \cdot \|\vec{b}\|}$$

e.g. (1,2), (4,6)
$\cos\theta = \frac{1 \cdot 4 + 2 \cdot 6}{\sqrt{1^2 + 2^2} \cdot \sqrt{4^2 + 6^2}} = \frac{16}{\sqrt{5}\sqrt{52}} \approx 0.992277$

(1,1,2,3) (2,4,1,5)
$\cos\theta = \frac{1 \cdot 2 + 1 \cdot 4 + 2 \cdot 1 + 3 \cdot 5}{\sqrt{1+1+4+9}\sqrt{4+16+1+25}} = \frac{23}{\sqrt{15}\sqrt{46}} \approx 0.87559$

5. Chebyshev distance
↳ greatest distance along any coordinate dimension

$$\text{Chebychev distance} = \max_i |x_{i1} - x_{i2}|$$

e.g. (1,2) (4,6)
$d = \max(3, 4) = 4$

(1,1,2,3), (2,4,1,5)
$d = \max(1, 3, 1, 2) = 3$

6. Jaccard index

↳ similarity score

↳ to compare the similarity and diversity of sample sets

A, B → sets

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

e.g. $A = \{1,2,3,4\}$, $B = \{3,4,5,6\}$

$$J(A,B) = \frac{2}{6} = 0.33$$

7. Hamming distance

$$\text{Hamming Distance} = \sum_{i=1}^{n} 1(x_i \neq y_i)$$

$$x_i \neq y_i = \begin{cases} 1 & \text{ith char doesn't match} \\ 0 & \end{cases}$$

e.g. 1100101, 1001001

```
1100101
1001001
-------
0101100  → d = 3
```

diff

k-NN

- It is a supervised classification algo. It relies on distance metrics to determine the closeness of data points.
- It predicts the class of a given point based on the majority class of k of its nearest neighbors.
- k is a positive integer, typically not large.
- If k=1, then the new data point is just assigned to the class of the nearest neighbor.

Steps

1. Choose the # of neighbors (k).
2. Calculate distances: of the new data point from all the data points in the training set.
3. Find the k nearest neighbors: use distance or similarity scores.

smallest k-values largest k-values

4. Vote for labels.

Choosing the right value of k

- Small k: makes the model sensitive to noise. The decision boundary is more intricate.
- Large k: makes the boundary smoother and the model more resistant to outliers. It'll blur the class distinctions.

Choosing the distance metric

- Euclidean: good for low-dim data.
- Manhattan: good for high-dim data, or for data with underlying grid-like structure.
- Cosine: useful when magnitude of data vectors is irrelevant.
- Jaccard index: when a feature contains sets as data.
- Hamming distance: when a feature contains strings of equal length as data.

e.g.

Instance	x_1	x_2	y
1	1	3	0
2	2	2	0
3	3	1	0
4	6	1	1
5	7	2	1
6	8	3	1

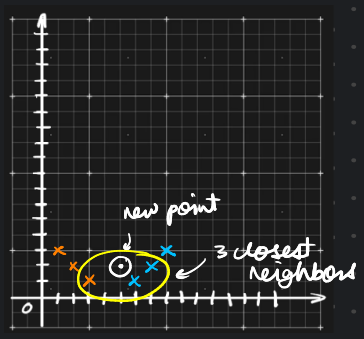
calculate the predicted label ($\hat{y}$) for $x = (5, 2)$.
use KNN with $K = 3$.

→ using Euclidean distance,

$d(x_1, x_{new}) = \sqrt{4^2 + 1^2} = \sqrt{17}$
$d(x_2, x_{new}) = \sqrt{3^2 + 0^2} = 3 = \sqrt{9}$
$d(x_3, x_{new}) = \sqrt{2^2 + 1^2} = \sqrt{5}$
$d(x_4, x_{new}) = \sqrt{1^2 + 1^2} = \sqrt{2}$
$d(x_5, x_{new}) = \sqrt{2^2 + 0^2} = \sqrt{4}$
$d(x_6, x_{new}) = \sqrt{3^2 + 1^2} = \sqrt{10}$

sorting: $\sqrt{2}, \sqrt{4}, \sqrt{5}$
$x_4 \quad x_5 \quad x_3$
$1 \quad 1 \quad 0$

by majority vote,
$\hat{y}_{new} = 1$



A	1	1	1	Blue
B	1	3	2	Blue
C	4	2	5	Green
D	5	3	2	Green
E	6	1	1	Red
F	2	2	2	Red
G	3	3	3	Red
H	4	1	2	Green
I	2	1	3	Red

Classify a new data point with features ($X_1 = 3$, $X_2 = 2$, $X_3 = 2$) using K-Nearest Neighbors with $K = 3$ using the following dist metrics :- (i) Euclidian (ii) Manhattan (iii) Chebyshev (iv) Cosine Similarity

→

$\|x_{new}\| = \sqrt{9+4+4} = \sqrt{17}$

		euc. dist from (3,2,2)	man dist	chebyshev dist	$\|\vec{x_i}\|$	$x_i \cdot x_{new}$	$\frac{x_i \cdot x_{new}}{\|x_i\| \, \|x_{new}\|}$
x_1	(1,1,1)	$\sqrt{2^2+1^2+1^2} = \sqrt{6}$	2+1+1 = 4	2	$\sqrt{3}$	7	0.98019 ←
x_2	(1,3,2)	$\sqrt{2^2+1^2+0^2} = \sqrt{5}$	3	2	$\sqrt{14}$	13	0.84266
x_3	(4,2,5)	$\sqrt{1^2+0^2+3^2} = \sqrt{10}$	4	3	$\sqrt{45}$	26	0.94003
x_4	(5,3,2)	$\sqrt{2^2+1^2+0^2} = \sqrt{5}$	3	2	$\sqrt{38}$	25	0.9836 ←
x_5	(6,1,1)	$\sqrt{3^2+1^2+1^2} = \sqrt{11}$	5	3	$\sqrt{38}$	22	0.8655
x_6	(2,2,2)	$\sqrt{1^2+0^2+0^2} = \sqrt{1}$ ←	1 ←	1 ←	$\sqrt{12}$	14	0.98019 ←
x_7	(3,3,3)	$\sqrt{0^2+1^2+1^2} = \sqrt{2}$ ←	2 ←	1 ←	$\sqrt{27}$	21	0.98019 ←
x_8	(4,1,2)	$\sqrt{1^2+1^2+0^2} = \sqrt{2}$ ←	2 ←	1 ←	$\sqrt{21}$	18	0.9526
x_9	(2,1,3)	$\sqrt{1^2+1^2+1^2} = \sqrt{3}$	3	1 ←	$\sqrt{14}$	14	0.9074

k = 3.

Using Euclidean dist.

x_6	x_7	x_8
R	R	G

Red

manhattan

x_6	x_7	x_8
R	R	G

Red

chebyshev

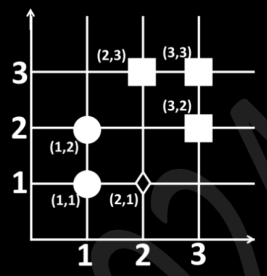
Red

cosine sim

B	R	R	G

Red

Q.63 Given the two-dimensional dataset consisting of 5 data points from two classes (circles and squares) and assume that the Euclidean distance is used to measure the distance between two points. The minimum odd value of k in k-nearest neighbor algorithm for which the diamond (◇) shaped data point is assigned the label square is _____.



2024

- ○ $x_1 = (1,1)$
- ○ $x_2 = (1,2)$
- □ $x_3 = (3,2)$
- □ $x_4 = (3,3)$
- □ $x_5 = (2,3)$

dist. from (2,1)

$\sqrt{1}$ ← ← ←
$\sqrt{2}$ ← ← ←
$\sqrt{2}$ ← ←
$\sqrt{5}$ ←
$\sqrt{4}$ ←

k=1 → 0
k=3 → 0
k=5 → 0

(S)

Q. Suppose we have a dataset of news articles with their content tokenized into sets of keywords.
Classify a new article with the keywords {econ, stock, bond} using k-NN (k=3). Use Jaccard index as the distance metric.

Article_ID	Keywords	Category
1	{politics, econ, govt}	Politics
2	{stock, econ, market}	Business
3	{govt, tax, budget}	Politics
4	{music, album, tour}	Entertainment
5	{bond, finance, investment}	Business

S = {economy, stock, bond}

	JI
A1	1/5 = 0.2 ← politics
A2	2/4 = 0.5 ← business
A3	0
A4	0
A5	1/5 = 0.2 ← business

(since JI is a similarity index, we take greater values)

for K=3, $\hat{y}_{new}$ = business

Q. Suppose we have a dataset of DNA sequences from different species, each represented by a string of nucleotides of equal length. We need to classify a new DNA sequence 'ACCCT' using k-NN with k=3 and hamming distance.
Dataset:

Seq_ID	DNA Sequence	Species
1	AACGT	X
2	AAGGT	X
3	AACCT	Y
4	CCCGT	Y
5	AAGTT	Z

	Hamming dist
S1	2 ←
S2	3
S3	1 ←
S4	2 ←
S5	3

$\hat{y} = Y$